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Reducing Emissions in Road Construction Rubble Transport

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Abstract

We present our entry to a competition in Norway centered around reducing Greenhouse gas emissions stemming from the transport of rubble at road construction sites. To this end, we were provided data collected by a Norwegian construction company, relating to the machines operating at some of their construction sites. Based on this data, we present some of our initial observations on the nature of this data, and present hypotheses as to how one might (a) optimize the flow of traffic to- and from a rubble loading site, such that trucks arrive when and only when needed, and (b) arrive at an optimal driving scheme between a rubble loading- and dump site.

Introduction

The "RoadAI – Reducing emissions in road construction" competition[1] was hosted by the Norwegian Artificial Intelligence Research Consortium (NORA), in collaboration with Skanska Norge AS, the Norwegian subsidiary of a Swedish multi-national construction company.

The onset for this competition has been that roughly 1.5% of Norwegian Greenhouse gas emissions come from construction machines[2]. It is worth noting that this number includes both work on site, and rubble transport. However, the Norwegian landscape, characterized by a mountainous, rocky terrain, suggests that a great deal of those emissions stem from transporting stone.

In the interest of reducing these emissions, Skanska has been collecting data relating to the work of their rubble transport machines (i.e., dumpers and dump trucks) at some of their construction sites. This includes frequent geolocation (GPS) and (less fine-grained) fuel consumption data, as well as drone footage of the construction sites. A fragment of this data has been provided as a dataset for this competition.

To those ends, we begin with a characterization of the load and dump sites, as this isn't provided to us in a reliable or wholesome manner in the handed out dataset. We then create a model for how quickly rubble is generated at a loading site. This gives us grounds for an optimal schedule for trucks to arrive, and leave the loading site fully loaded. Finally, by factoring in elevation data from the Norwegian Mapping Authority, we also provide an optimal driving scheme for when the truck leaves a loading site, and heads for the dump site.

Automatic Detection of Load and Dump Sites

Although the handed out data has the fields Load- and Dump- -Latitude and -Longitude, these data points rely on feeble human input. Hence, we need an automatic procedure for picking points to serve as indicators of these sites. Furthermore, a mere collection of latitudes and longitudes is not a good characterization of a geographic region. Some simple[1] alternatives, are to find a bounding box, or a convex hull around the points in question.

Before finding suitable collections of points which to use for the dump and loading sites, we illustrate the relationship between various kinds of machines by drawing bounding boxes around their locations (see Figure 1). We observe that, in general, dumpers operate within a smaller region than corresponding trucks. This observation coincides with the provided information that "loading sites are where dumpers and trucks intersect".

We also observe that dumpers of Equipment and Soil do not interfere, while Stone dumpers may interfere with the other dumpers. Finally, although all truck types may interfere as they move to and from their loading sites, Equipment and Soil trucks operate in a much smaller region than do Stone trucks. [2]

[1] We do not consider more complex polytopes here.

[2] See the provided "Explore GPS Data" Notebook for an illustration of convex hulls, yielding similar observations.

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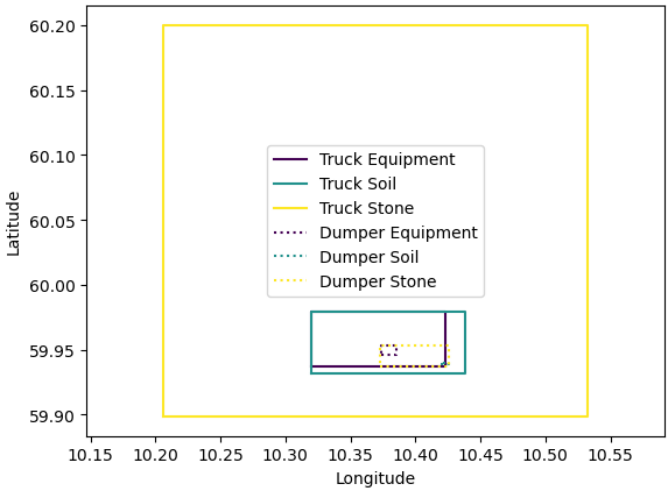


Figure 1: Bounding boxes around the GPS data points machines carrying different types of material (03-07-2022)

Load Sites

It follows from the above observations that it is best to segregate loading sites by material, and that we can characterize a loading site by drawing a convex hull around the positions of the dumpers of that material. See Figure 2 for an illustration of this approach.

Dump (and Load?) Sites

It is perhaps safe to assume that trucks will spend a considerable amount of time at both a load and dump site–it would make sense aim to spend as little time as possible at any point between the two. Hence, to find both load and dump sites (but following the above, dump sites in particular), we can use a density-based clustering algorithm. That is, to identify regions where many GPS coordinates occur.

DBSCAN[3] is one such algorithm. We illustrate our use of DBSCAN on a single trip in Figure 3. This method clearly identifies a region at both ends of the travel path. Using DBSCAN however, requires some careful parameter tuning. Trucks also spendconsiderable amounts of time at road junctions, and may be slowed down by traffic.

Which of the two sites identified by DBSCAN is the load and dump site can be determined by intersecting with the loading site found using the method mentioned above, or by considering the timestamps of the data points at those locations—a trip begins and ends at loading site (see also Figure 4).

Modelling Loading Site Rubble Generation

In order to be able to schedule for trucks to arrive at a loading site only when they are needed, and leave fully loaded, we need to model how quickly rubble is generated at the loading site.

We can do this by considering historical data of truck entry times at the loading sites, and exit times, coupled

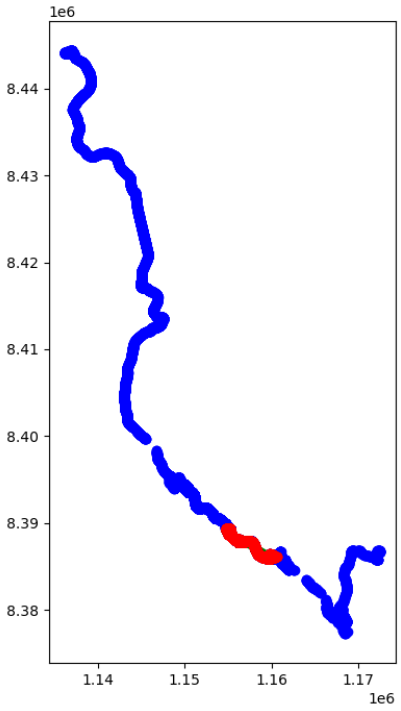


Figure 2: Stone trucks outside dumper convex hull (blue) and trucks inside dumper convex hull (red) in epsg:4326 units (i.e., km) (03-07-2022)

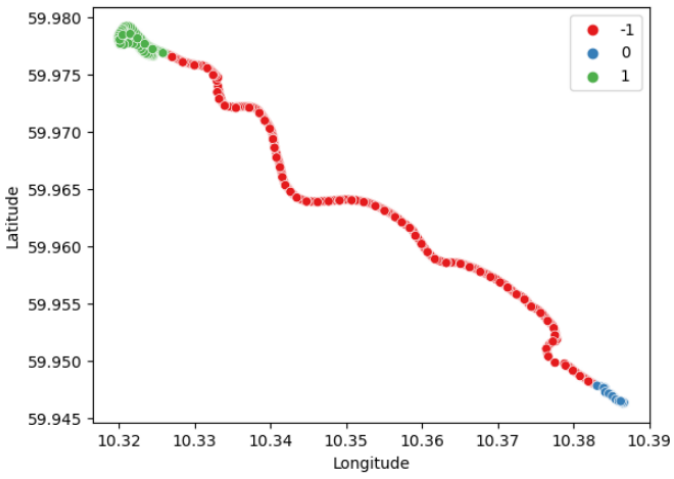


Figure 3: DBSCAN of truck trip (03-07-2022, 6225a07a03d57f0000de776a) with parameters $\epsilon = 0.001$ and min_samples $= 15$. -1 (red) stands for points outside of high-density regions.

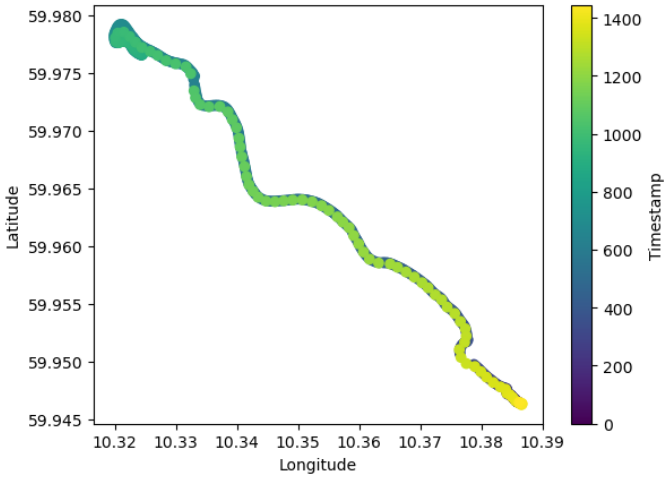


Figure 4: Timestamp lat/lon map of trip (03-07-2022, 6225a07a03d57f0000de776a)

with how much mass they carried out. Furthermore, it is perhaps safe to assume a linear model of rubble generation for any given truck.

The speed of stone generation for a given day is illustrated in Figure 5. For details about the algorithm, see the function `by_minute` in the submitted "Trip patterns" Notebook.

Future Work

Armed with a model of how quickly stone is generated, we can now schedule adequately sized trucks to arrive only when they are needed. To further optimize the driving scheme from the loading site to dumping site (and back), we need to consider the elevation at each position along the path.

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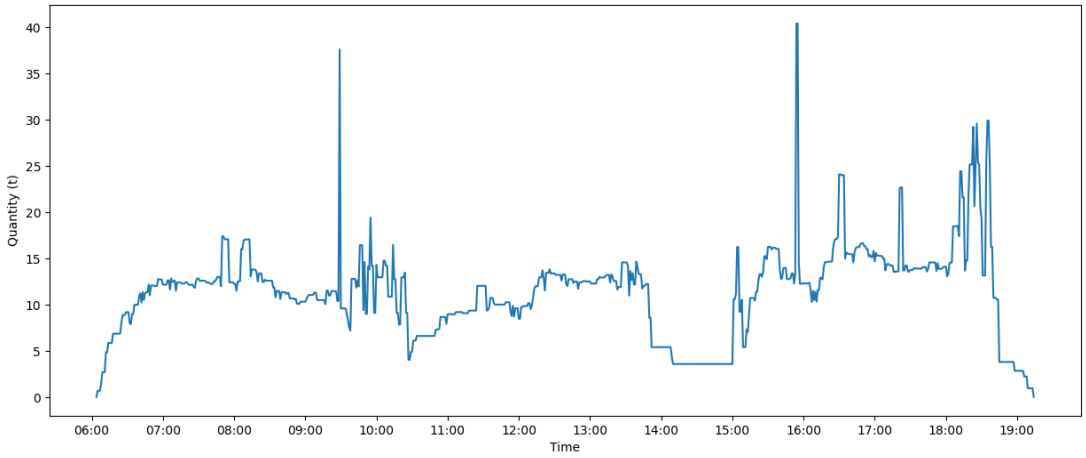


Figure 5: Stone generation at loading site, by the minute (03-07-2022)